The Potential for REDD+: Key Economic Modeling Insights and Issues

Ruben N. Lubowski* and Steven K. Rose†

Introduction

Forestry and agricultural land uses contribute roughly equally to produce approximately a quarter of global greenhouse gas (GHG) emissions, chiefly in developing countries, and they are projected to continue as significant emission sources in the absence of new policies (Rogner et al. 2007). Although the Kyoto Protocol excluded mechanisms for reducing deforestation, the Cancun Agreements of December 2010 affirmed that activities in developing countries for reducing emissions from deforestation and forest degradation, in addition to maintaining and increasing carbon stocks through other forestry activities (REDD and REDD+, respectively), should form part of any future global climate agreement, and they called for the exploration of public and market-based approaches to financing REDD+. The basic idea is that REDD+ would provide payments to jurisdictions (i.e., countries, states, or provinces) that voluntarily reduce forest emissions below agreed-upon benchmark levels (United Nations Framework Convention on Climate Change [UNFCCC] 2010). The governments of Norway and some other industrialized countries are already providing funding to build institutional capacity and begin implementing REDD+ on a pay-for-performance basis. Most notably, Brazil has implemented policies that have reduced deforestation more than 75 percent since 2004, in part through a $1 billion agreement with Norway (Nepstad et al. 2009; Soares-Filho et al. 2010). Other existing REDD+-related activities include forestry credits being sold in voluntary carbon markets to individuals, corporations, and investors (see Peters-Stanley, Hamilton, and Yin 2012), and policymakers designing approaches for REDD+ use within compliance markets, where GHG emissions permits are available to firms, subject to regulatory

*Chief Natural Resource Economist, International Climate Program, Environmental Defense Fund, Washington, DC; e-mail: rlubowski@edf.org.

†Senior Research Economist, Energy and Environmental Analysis Research Group, Electric Power Research Institute, Washington, DC; e-mail: srose@epri.com.

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constraints, or caps, on aggregate emissions (e.g., California's cap-and-trade system, which began permit auctions in late 2012).

This article, which is part of a symposium on the economics of REDD,[1] reviews the main findings and insights from economic modeling of REDD+, examines the limitations of these models, and discusses key issues and priorities for future research on REDD+. Economic modeling has indicated low opportunity costs and significant near-term mitigation (emissions reduction) potential through reducing deforestation and undertaking afforestation and other activities to enhance forest carbon stocks, which together could substantially lower net carbon dioxide (CO_2) emissions and play an important role in global climate change policies. However, until recently, most modeling has not addressed implementation and policy design issues. Recent analyses of implementation uncertainties and challenges suggest a more limited and nuanced mitigation role for REDD+, especially in the near future. These recent insights, as well as modeling challenges, suggest that the actual costs and environmental benefits of REDD+ are uncertain and highly dependent on policy and implementation features.

The next section presents a brief overview of REDD+ applications and modeling tools. This is followed by a review of the main findings, insights, and limitations of these models. We discuss both early studies, which generally make idealized, and hence optimistic, assumptions about REDD+ policies and institutions, and a more recent strand of the literature that explores the real-world complexities of policy design and implementation, which affect the cost, net benefit, and distributional impacts of REDD+ programs. We then identify REDD+ research needs and challenges. The final section presents a summary and conclusions.

REDD+ Applications and Economic Modeling Tools

Decisions under the UNFCCC define REDD+ as "policy approaches and positive incentives on issues relating to reducing emissions from deforestation and forest degradation in developing countries; and the role of conservation, sustainable management of forests and enhancement of forest carbon stocks in developing countries" (UNFCCC 2010). Because economic studies do not typically include reducing degradation, in this article, REDD+ refers to reducing deforestation plus at least afforestation or reforestation (planting trees on lands cleared of forests in the recent past). We refer explicitly to reducing or avoiding emissions from deforestation when this is the only activity considered. For consistency with most studies, we do not distinguish between *reducing* emissions below current levels and *avoiding* future increases in these emissions, although the latter is defined as forest "conservation" by the UNFCCC.

[1]The other articles are Kerr (2013), which introduces the symposium and focuses on the economics of international policy to reduce emissions from deforestation and degradation; Angelsen and Rudel (2013), which uses the concept of a forest transition to examine the feasibility and effectiveness of REDD policies; and Pfaff, Amacher, and Sills (2013), which examines economic theory and empirical evidence concerning the impacts of domestic policies on forest loss.

REDD+ Applications

Economic modeling has generally been used to inform four types of REDD+ applications that are relevant to climate change science and policy:

(1) Site-specific planning: spatially explicit land use scenarios for site-specific evaluation of potential REDD+ activities including estimates of the likelihood of deforestation and the potential for avoiding it on specific land parcels;[2]
(2) REDD+ supply potential: estimated costs of producing increasing quantities of emissions reductions through REDD+ activities for countries, regions, and the world;
(3) Climate management potential: estimates of the potential role of REDD+ in long-run multisectoral management of climate change (e.g., to limit atmospheric GHG concentrations or target future radiative forcing or temperature change);[3] and
(4) GHG market potential: estimates of the potential role of REDD+ in existing and potential GHG permit markets and abatement activity portfolios for compliance with national or regional emissions reduction targets over time.

The majority of studies have produced optimistic benchmarks for each application that do not consider policy realities and practical implementation issues. More recent analyses have focused on informing the specific policy design and implementation of REDD+ and assessing implications for the different types of REDD+ applications. This topic is discussed in detail later.

Economic Modeling Tools

The literature is filled with a diverse set of economic modeling tools that reflect the various REDD+ applications and varying scales. There are spatially detailed models at the local to country scale, aggregated models at national, regional, and global scales, and hybrids. There are four classes of REDD+ economic models: fixed market, partial equilibrium, general equilibrium, and integrated assessment (Table 1). The classes are listed in order of increasing level of endogeneity (market and climatic).[4] These classes also generally reflect increasing levels of scope (spatial and temporal) and decreasing levels of detail, and they loosely align with the four application types listed above.

Models differ in terms of the specificity of the forestry activities for reducing carbon emissions or increasing carbon stocks. Some models use more specific bottom-up characterizations of management details and costs, including forestry details such as regional tree species, age cohorts, timber rotations, and other management options (e.g., Sohngen and Sedjo 2006). Other models characterize generic top-down technologies that are considered representative or average (e.g., Wise et al. 2009). Models also differ in the spatial resolution of decision making, from fine-scale geographic units such as individual cells within a gridded landscape

[2]Sometimes these estimates are used to justify particular projects under existing voluntary carbon market protocols.
[3]Radiative forcing is a measure of the difference between radiant energy received by the earth and energy reradiated to space. Greenhouse gases and other factors (e.g., land reflectivity and aerosols) can affect this difference and change the planet's climate.
[4]That is, fixed market models consider neither changes in output and input prices nor changes in climate, but integrated assessment models consider both.

Table 1 Models used for economic analyses of REDD+

Modeling type	Economic scope	Resolution (spatial, sectoral, temporal)	Market endogeneity	Examples
Fixed market	Site(s) specific (e.g., a forest(s), parcel, grid)	High	None	Benítez and Obersteiner (2006); Nepstad et al. (2007)
Partial equilibrium	Sectoral and longer-term (e.g., forests and/or agriculture, 50–100 years)	Moderate (e.g., forest type and age cohorts, some land heterogeneity)	Sector output, some input prices and quantities (e.g., land), sometimes demand price response	Busch et al. (2009, 2012); Havlik et al. (2010); Rose and Sohngen (2011); Sathaye et al. (2006); Sohngen and Sedjo (2006)
General equilibrium	Economy-wide and nearer-term (e.g., 20 years)	Moderate (e.g., alternative land-using sectors, limited forest resolution, some land heterogeneity)	All output and input prices and quantities, demand price response	Cattaneo (2001); Golub et al. (2009); Gurgel, Reilly, and Paltsev (2007);
Integrated assessment	Linked global economy and biophysical systems and very long-run (e.g., at least 100 years)	Low to moderate (e.g., linked with land use sectoral models, or using sectoral supply schedules, aggregate commodity sectors)	Output and some input prices and quantities, demand price response	Bosetti et al. (2011); Riahi, Gruebler, and Nakicenovic (2007); Reilly et al. (2012); Tavoni et al. (2007); Wise et al. (2009)

(e.g., Rokityanskiy et al. 2007; Soares-Filho et al. 2010) to decisions at the subnational, national, or global regional levels, although they might use fine-scale geographic data and report spatially downscaled results (e.g., Gurgel, Reilly, and Paltsev 2007).

Fixed market and partial equilibrium models

Fixed market models tend to have a narrow spatial and temporal focus and therefore can assume exogenous market conditions. Partial equilibrium models capture changes in forestry and agricultural commodity prices and domestic and international demand responses, as well as long-run investment decisions. Therefore they are suited for estimating potential large-scale implications of changes in land use incentives across regions, which are likely to occur with regional or global adoption of REDD+. For example, reductions in deforestation could lower the supply of soybeans, cattle, and/or timber, raising their prices and therefore both the costs of REDD+ and the risks of shifting deforestation elsewhere (known as leakage). Because partial equilibrium models focus on large-scale changes and market implications, their spatial and temporal resolution is coarser than in fixed market models. Some partial equilibrium models incorporate long time horizons in order to capture the effects of expectations about future markets on current forest and land use decisions (e.g., Sohngen and Sedjo 2006). Partial equilibrium models have been used extensively in projecting potential net regional carbon emissions from forests over time periods ranging from the next few decades to a century, and under different scenarios, including estimating REDD+ cost schedules (e.g., Rose and Sohngen 2011; Sathaye et al. 2006; Sohngen and Sedjo 2006). Such models have also been applied to analyze the efficiency, effectiveness, and equitability of REDD+ rules and policy incentives (e.g., Busch et al. 2009, 2012; Cattaneo et al. 2010).

General equilibrium models

General equilibrium models can capture land use competition and market interactions with non-land sectors, input and economywide output price responses, household budget and total consumption implications, and bilateral trade responses. These models make it possible to analyze cost-effective GHG mitigation across the economy, as well as a broad suite of REDD+ policy levers, including those with indirect implications for the drivers of deforestation. For example, national general equilibrium models have been used to analyze the deforestation effects of exchange rates, land tenure, agricultural technologies, and transport costs in Brazil (Cattaneo 2001) and production forestry subsidies in Indonesia (Warr and Yusuf 2011). These types of models have also been used to estimate the cost implications of including REDD+ credits in potential US federal cap-and-trade policies (US Environmental Protection Agency [EPA] 2010). In addition, general equilibrium models have been used to examine the impacts of global versus regional mitigation incentives on regional forestry and agricultural mitigation supplies (Golub et al. 2009).

Integrated assessment models

Integrated assessment models (IAMs) are the most comprehensive economic models used for climate policy analyses, and the most aggregated in terms of markets, regions, and time periods. They model global economic activity and biophysical processes (terrestrial, oceanic, and atmospheric systems), and are designed to simulate alternative socioeconomic and technological scenarios and the resulting future climate. Some IAMs include limited types of climate feedbacks on the economy, such as changes in crop and forest productivity (e.g., van Vuuren et al. 2006). IAMs use a variety of approaches to include avoided deforestation and other land use activities (with varying degrees of sophistication and endogeneity), and some draw on separate forest and specific land use models (e.g., Reilly et al. 2012; Rose et al. 2012; Wise et al. 2009).

The next three sections discuss three categories of insights from economic modeling of REDD+: (1) REDD+ supply estimates under idealized assumptions; (2) GHG management implications of the idealized supply estimates (i.e., REDD+ quantities resulting from the interaction of supply and demand for meeting a particular climate objective); and (3) REDD+ implementation and policy design challenges and implications for REDD+ supplies and climate management potential under more realistic assumptions.

Estimates of the Supply of REDD+ under Idealized Assumptions

The starting point for virtually all economic evaluations of the potential role of REDD+ in climate policy is an assessment of the costs per unit of emissions reductions. For example,

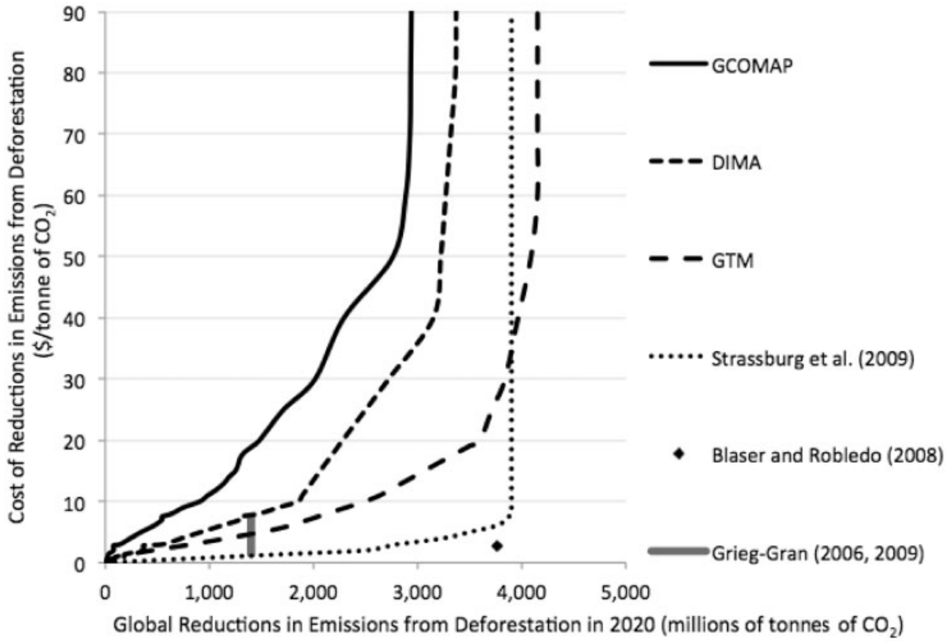

Figure 1 Global supply curves and point estimates for reducing emissions through avoiding deforestation (for 2020)

Notes: Estimates from three global sectoral models—the Generalized Comprehensive Mitigation Assessment Process Model (GCOMAP), the Global Timber Model (GTM), and the Dynamic Integrated Model of Forestry and Alternative Land Use Model (DIMA)—are from the review in Kindermann et al. (2008).
Source: Adapted from Murray, Lubowski, and Sohngen (2009).

Figure 1 displays various estimates of the potential global supply of avoided carbon emissions from deforestation in 2020. Other forest carbon sequestration strategies are discussed later. Although the estimates vary greatly, they all suggest that the potential for reducing emissions through avoided deforestation is massive and inexpensive. However, these studies generally assume idealized REDD+ circumstances, and thus they likely overestimate the actual potential to reduce deforestation emissions. Specifically, they assume that an incentive to protect and/or increase forest carbon stocks is available immediately and comprehensively, and they ignore implementation challenges. These issues are discussed in the section 'Considering Implementation Realities.'

Reduced Deforestation Opportunity Cost Estimates

Most REDD+ cost estimates, including some of the estimates in Figure 1, focus on the opportunity costs of changing land use from the perspective of a private land owner or other user, that is, the net benefits foregone due to a shift away from the most profitable land use option, such as growing soybeans or ranching cattle. These costs can be thought of as the price of compensating a landowner or land user for reducing carbon emissions or increasing carbon sequestration.

Reduced deforestation bottom-up estimates

The majority of opportunity cost estimates are for reducing deforestation and based on bottom-up studies conducted at a regional scale using detailed data regarding both returns to alternative land uses and carbon stocks per hectare. In general, these analyses yield the lowest opportunity cost estimates, that is, the lowest price per tonne of CO_2 emissions reduced for a given quantity of reduced deforestation. Boucher's (2008) review of twenty-nine studies finds an average opportunity cost for reducing tropical deforestation of \$2.51/tonne CO_2 (tCO_2), with eighteen studies reporting costs below \$2 and all but one below \$10. These studies usually report point estimates of average costs for reducing a given amount of deforestation in a particular region. Some studies (e.g., Blaser and Robledo 2008; Grieg-Gran 2006, 2009; shown in Figure 1) have extrapolated these results to a global scale and generated low cost point estimates for major reductions in global deforestation (65 percent and 50 percent, respectively).

Bottom-up methods have also been used to estimate regional and global supply curves (versus point estimates) for deforestation emissions reductions. Strassburg et al. (2009) (presented in Figure 1) is a global example. Nepstad et al. (2007), a regional example, estimated opportunity costs of \$0.76/tCO_2 for eliminating deforestation in 94 percent of the Brazilian Amazon over ten years, with costs doubling when protection was extended to the remaining lands where high return soybean production rather than ranching must be displaced.

Bottom-up studies are particularly useful for examining and comparing the potential for REDD+ in specific locations. However, their estimates of regional and global potential should be viewed with caution because they ignore market effects and interactions, including leakage, which are likely to occur with large-scale implementation of REDD+.

Reduced deforestation sectoral estimates

In addition to bottom-up estimates, Figure 1 includes estimated supply curves for reduced deforestation emissions for 2020 from three widely used global sectoral models, as reported in Nabuurs et al. (2007) and Kindermann et al. (2008). These models include two forest sector partial equilibrium models, the Generalized Comprehensive Mitigation Assessment Process Model (GCOMAP) and Global Timber Model (GTM); and a global forestry model (Dynamic Integrated Model of Forestry and Alternative Land Use [DIMA]) that projects land responses to satisfy specified commodity demands over time. Kindermann et al. (2008) found that differences across these three modeling frameworks and the underlying data, including carbon densities and opportunity costs, led to different projected baseline deforestation rates and therefore avoided deforestation supply estimates (e.g., 1.3 to 3.0 billion tonnes of CO_2 at \$15/tCO_2). Despite the variation, all three global sectoral model results suggest that avoided deforestation is substantially more expensive than the bottom-up estimate of Strassburg et al. (2009) and most point estimates. These lower estimates of REDD+ supply reflect, at least in part, the fact that they account for commodity market feedbacks.

The global sectoral studies also provide insights regarding the potential regional distribution and dynamics of REDD+ supplies. In Kindermann et al. (2008), the DIMA and GCOMAP models project the largest potential for mitigating deforestation emissions in Africa and Latin America; the GTM model projects greater relative potential in Latin America and Southeast Asia. All three models suggest Africa could be important in terms of high deforestation emissions as well as relatively low costs to avoid them.

Increased Carbon Sequestration from Other Forestry Activities

Although the studies shown in Figure 1 highlight the low-cost potential to reduce deforestation emissions in the coming decades, afforestation, reforestation, and other changes in forest management, including changes in rotations, species mix, and management intensity, have even greater estimated potential over the long term. Rose et al. (2012) find that the bulk (78–99 percent) of global cumulative forest carbon gains estimated by two of the sectoral models occurs between 2050 and 2100. The majority of these gains occur in the tropics, but tropical carbon gains are eventually dominated by forestation activities, rather than reduced deforestation. A significant share (33–43 percent) of the potential forest carbon gains also occurs in temperate regions, where forest management activities provide more than half of the net reductions. Delays in policies to reduce deforestation could, however, increase the relative supply from afforestation, reforestation, and forest management, and decrease the supply of reduced deforestation in tropical regions (Rose and Sohngen 2011). We elaborate on this point when we discuss implementation issues and challenges.

Implications for Greenhouse Gas Management

While the supply estimates convey information about the costs of achieving different quantities of REDD+, policymakers are interested in the contribution, and value, of REDD+ as one strategy for addressing climate change. As discussed next, climate management (stabilization) studies have estimated the role of REDD+ activities in stabilizing atmospheric GHG concentrations or in achieving long-run radiative forcing targets. GHG market studies, in contrast, have estimated the role of REDD+ credits within competitive GHG emissions permit markets based on emissions caps. Both sets of analyses are cost-effectiveness studies that estimate the potential cost savings from including REDD+ mitigation in the suite of eligible mitigation technologies to achieve particular climatic objectives. This literature has generally assumed perfect information but has also considered the potential for REDD+ to hedge against market and technology uncertainties and reduce future market price volatility.

Climate Management Studies

Climate management studies have examined the role of some or all of the following REDD+ activities: avoided deforestation, afforestation, reforestation, and other forest management. Forest management is the most difficult to model as it requires dynamic tracking of forest inventories. Because studies differ in terms of the REDD+ activities modeled, as well as other key factors, including the electric, industrial, and transportation sector technologies, baseline assumptions regarding economic growth, population, and technological progress, climate objectives modeled, and basic modeling structure and algorithms, it is difficult to compare results directly. Together, these model features and assumptions define the potential cost-effective role of reduced deforestation and the broader suite of REDD+ activities. Many studies have considered other land-based mitigation strategies as well, such as biomass for energy and management of agricultural GHG emissions and soil carbon (for example, see Rose et al. 2012).

For this article, we reviewed numerous climate management studies and found estimates that REDD+ activities could cost-effectively contribute between 4 and 55 percent of overall cumulative emissions reductions through 2050.[5] Thus reducing deforestation and other land-based abatement could play a small or significant role in achieving and reducing the costs of stabilization policies over the next century. The lower estimates generally come from studies that include afforestation/reforestation as the only forest carbon mitigation option available or that include substantial use of biomass-based energy, which could be a powerful competitor for land resources (Rose et al. 2012). The higher estimates generally come from studies that include reduced deforestation. However, the forest carbon and land use changes in these analyses are usually based on idealized policy assumptions, and sometimes external modeling that is not fully integrated with the economic and climatic processes being modeled.

Studies have also estimated the potential of global forestry mitigation under a theoretical economically optimal climate stabilization pathway that balances the benefits and costs of avoiding climate change (Sohngen and Mendelsohn 2003; Van't Veld and Plantinga 2005). They find that forestry activities lower the cost of mitigation but only slightly reduce the optimal level of mitigation because of the way climate change damages are modeled: marginal climate damages decline slowly with annual emission reductions. However, given large uncertainties regarding both the climate response and the impacts of climate change, most long-run climate management modeling has concentrated on estimating cost-effective mitigation portfolios for achieving an atmospheric GHG concentration or other climatic goal.

As a potentially readily deployable technology, REDD+ could also have additional value as a near-term hedging strategy against climate change uncertainty (Bosetti et al. 2011; Golub 2010). Early emissions reductions of any kind may have value as an insurance policy that preserves future options in light of scientific uncertainty (Anda, Golub, and Strukova 2009; Fisher et al. 2007). However, as discussed later, the amount and net benefit of REDD+ available in the near term is uncertain given implementation challenges. In addition, given climatic uncertainties, Gitz, Hourcade, and Ciais (2006) find it is desirable to delay afforestation/reforestation as these options can be deployed later if greater reductions are in fact needed. This hedging strategy of delaying forestation is in contrast to avoided deforestation, which presents a time-limited opportunity to avoid carbon losses.

Greenhouse Gas Market Studies

Studies of REDD+ in GHG emissions permit markets also suggest a significant opportunity to protect tropical forests while reducing the costs of climate policy by 25–40 percent in cases examining reduced deforestation only and by 40–65 percent in cases where other forestry activities, such as afforestation and reforestation, are also considered (see e.g., den Elzen et al. 2009; Murray, Lubowski, and Sohngen 2009).[6] In a GHG permit market, the estimated demand for REDD+ depends on the cap on regulated emissions, the availability and cost of other emission reduction alternatives, and the institutional rules for using REDD+ and for trading across sectors, regions, and time periods. Concerns about whether REDD+ credits represent climate benefits that are equivalent to reductions in sectors covered by emissions caps (e.g., whether REDD+ reductions are real and to what extent they increase emissions

[5]See the online supplementary materials for this article for our summary of studies and estimates.
[6]The individual estimates vary with differences in the supply and demand factors modeled. See the supplementary materials for a summary of studies.

elsewhere), the desire to promote domestic and energy sector mitigation, as well as other factors, have resulted in policy proposals that limit REDD+ and the use of other "offsets."[7]

An important difference between studies is whether they use static or dynamic modeling, with the latter approach allowing for trading of emission permits ("allowances") and GHG reduction credits over time. Single-period analyses are typical in European policy contexts (e.g., den Elzen et al. 2009; Eliasch 2008), which generally target near-term reductions, while dynamic approaches characterize the modeling of proposed US policies that include emission caps through 2050 (e.g., US EPA 2010; see Piris-Cabezas 2010 for more discussion). Static GHG market studies have estimated that unrestricted inclusion of credits from avoided deforestation could lower allowance prices by 10–45 percent and by 29–53 percent when other forestry activities are included, with smaller impacts when there are quantity restrictions (see, e.g., Eliasch 2008). Dynamic studies find that banking allows greater forestry mitigation overall with a less extreme range of price impacts (15–29 percent for avoided deforestation and 31–43 percent for REDD+) as credit use is spread more gradually over time (see, e.g., Murray, Lubowski, and Sohngen 2009).

Regulated entities have a potential incentive to save or "bank" emissions permits or emission reduction credits in excess of current obligations when future allowance prices are expected to be higher or are uncertain, which is likely with the anticipated tightening of emissions caps over time and technology, economic, and policy uncertainties. Modeling of carbon markets indicates that credit banking would be used heavily and even more so if REDD+ opportunities were available in the near term (e.g., US EPA 2010). In addition, banked credits and potential option contracts to buy future credits—based on REDD+ and other mitigation—could reduce the likelihood of high prices and help compliance entities hedge against carbon market volatility (Fuss et al. 2011; Kim 2010).

Overall, economic modeling under ideal implementation assumptions has suggested that including REDD+ activities within a GHG compliance market can significantly lower allowance prices and overall policy costs. It is important to note that these models also estimate that the majority of mitigation will still occur within the energy system.

We next examine two specific REDD+ GHG market issues that have been evaluated in the economics literature. First, concerns have been raised in policy debates about the possibility of REDD+ credits flooding the carbon market and reducing innovation, especially innovation of low-carbon energy technologies. Analyses of this issue (e.g., Bosetti et al. 2011; Tavoni, Sohngen, and Bosetti 2007) conclude that, if REDD+ is available and cost effective, it will, by definition, displace other higher cost technologies and result in a more efficient allocation of resources across the economy, which will increase welfare. This reallocation of resources could include increased investments in energy technology innovation. The flexibility provided by REDD+ could also help hedge technology development risks and grant time to businesses facing emissions permit obligations to optimize capital investments (Fuss et al. 2011).

Second, linking REDD+ to carbon markets has both potential distributional impacts and implications for global financial flows. Developing regions gain the value of REDD+ mitigation while industrialized countries benefit from lower mitigation costs. For example, Nepstad et al.

[7]For example, the rules for California's cap-and-trade system limit the use of international credits to offset regulated Californian emissions to 2–4 percent of the annual compliance obligations of regulated entities through 2020.

(2009) estimate that purchases of REDD+ for a US carbon market could generate revenues over time for Brazil valued at $37–$111 billion under different reference-level or baseline scenarios for crediting deforestation emissions reductions between 2013 and 2020. These revenues are enough to cover the costs of Brazil's national deforestation reduction target and potentially other mitigation. However, alternative reference levels for crediting suggest orders of magnitude differences in potential emissions reduction credits to particular countries (see, e.g., Griscom, Shoch, et al. 2009), which affects potential profits as well as incentives to voluntarily reduce emissions (Busch et al. 2009). Market prices and resulting producer and consumer surpluses from REDD+ trading will also depend on market constraints such as quantity limits on sales/purchases and trading ratios (e.g., Eliasch 2008), and whether buyers or sellers can exert market power (Murray, Lubowski, and Sohngen 2009).

Considering Implementation Realities

Thus far we have focused on studies that suggest that potential emissions reductions from avoided deforestation and other REDD+ activities are abundant, relatively low cost, and available in the near term on global scales. However, uncertainties (historical and future), missing costs, and the current state of institutions and international agreements raise questions about how and to what extent this potential can be realized in practice, especially in the near term. This section discusses several issues that qualify the results from previous modeling and highlight the importance of considering the realities of policy design and implementation.

Opportunity Cost Issues

First, current opportunity cost estimates may be incomplete and insufficient. Most modeling does not account for investment risk or transactions and administrative costs associated with design, contracting, monitoring, enforcement, and mitigation crediting. To the extent that REDD+ programs will involve private investments, there are significant investment risks. Recent studies indicate that accounting for institutional and political barriers reduces the mitigation potential. For example, Benitez et al. (2007) find that country risks (political, economic, and financial) could reduce REDD+ supplies by an estimated 60 percent, and Rose et al (2013) find that country and forest carbon project risks could reduce near-term global REDD+ supplies by 50 percent, primarily in developing and transitional regions.

Another opportunity cost issue is that cost estimates for a landowner may not reflect the indirect off-site economic costs to a country or region, such as losses to valued-added processing industries (Pirard 2007). However, opportunity cost estimates may not consider local legal or cultural restrictions that constrain land use alternatives and would imply lower opportunity costs in practice (Gregersen et al. 2010; Nepstad et al. 2007).

Furthermore, opportunity costs do not include the costs of developing institutional capacity and actually implementing and transacting a REDD+ program. Some researchers have estimated possible implementation costs—including $4 billion to build capacity and reform policies in forty countries (Eliasch 2008), $233–500 million per year to administer payments for environmental service approaches (Grieg-Gran 2009), and average transactions costs of $0.38/$tCO_2$ across eleven forestry mitigation projects (Antinori and Sathaye 2007). These estimates consider only a limited range of potential policies for implementing REDD+. Implementation of REDD+ may also be constrained by other social concerns, such as food security, local employment, and leakage.[8]

Finally, it is important to distinguish opportunity costs from the budgetary or financial costs that a government or other administrator would incur to achieve REDD+ (see White and Minang 2011). Budgetary costs could be negative, if there are emissions tax revenues, or exceed opportunity costs, depending on how payments are targeted (Börner and Wunder 2008; Kindermann et al. 2006). The incremental opportunity costs of REDD+ are also not likely to be the price a buyer would pay in a market setting. A buyer would pay the equilibrium market price, which would be set by the cost of the marginal mitigation alternative and would entail profits for all suppliers that can deliver emission reductions at lower costs. While opportunity costs may be useful for estimating the minimum societal cost for REDD+, the actual financial costs will reflect a complex set of economic, political, and institutional realities. Financial payments could entail transfers of resources within society (rather than economic costs to society as a whole) and could also have important distributional consequences that will be critical for policy design.[9]

Policy Design and Implementation Issues

The second issue is that the global REDD+ supply studies described earlier make a strong idealized policy assumption. They assume that a global comprehensive forest carbon policy will be implemented instantly, meaning that all countries will be willing and able to administer a program perfectly and immediately to preserve and enhance forests carbon stocks. In reality, access to global forest carbon supplies is unlikely to be global (which raises concerns about regional leakage), comprehensive (which raises concerns about possible leakage and other market interactions across REDD+ activities), or immediate (which raises concerns about the implications of delay and temporal coordination).

To explore this issue, Rose and Sohngen (2011) examine alternative near-term forest carbon policies that are incomplete (e.g., including only afforestation or only developed countries) or delayed as it will take time to implement a global forest carbon policy. They find that when policies do not include incentives for developing countries to avoid deforestation, the cumulative net forest carbon gains could actually be negative in the near term. Under all scenarios, deforestation accelerates in the near term as landowners seek to maximize long-term timber and carbon profits. Overall, the long-run cumulative potential for avoiding deforestation diminishes substantially. Rose and Sohngen (2011) also find competition and complementarities between forest carbon activities, with afforestation incentives increasing the cost of avoided deforestation (and vice versa), and forest management carbon incentives increasing the value of afforestation (and vice versa). To date, most supply and market analyses ignore these interactions within a period and over time.

The geographic scope of incentives to reduce emissions also affects the potential for both leakage and mitigation within and across countries. Murray, McCarl, and Lee (2004) study shifts in forestland and timber production within the United States and estimate leakage to range from less than 10 percent to more than 90 percent, depending on the type of forest carbon

[8]See Fisher et al. (2011) for an analysis of this issue in Tanzania.
[9]These distributional consequences also apply to transfers across countries.

activity and the region. Golub et al. (2009) find that multilateral climate policy action reduces international emissions leakage but raises the costs of mitigating US forestry and agricultural emissions as it creates a comparative trade advantage for US farmers because US agriculture is relatively less land extensive. Gan and McCarl (2007) find that under less than global conservation policies, international deforestation leakage can be as high as 42–95 percent in the forest products industry.

A recent literature also finds that policies to reduce emissions from fossil fuels could increase deforestation emissions because demand for bioenergy crops will expand if there are not parallel incentives to reduce emissions from land use (e.g., Reilly et al. 2012; Wise et al. 2009). Recent mandates for the use of biofuels in the United States and Europe have spurred analyses of the potential impacts of these policies on commodity prices and land use changes in other countries (e.g., Roberts and Schlenker forthcoming; Searchinger et al. 2008). The estimated impacts on deforestation are sensitive to various factors including forest conversion costs (Gurgel, Reilly, and Paltsev 2007) and acreage, yield, and trade responses (Keeney and Hertel 2009; Villoria and Hertel 2011).

The studies discussed earlier concerning REDD+ supplies and their potential role in climate policies also do not reflect the economic incentives created under likely REDD+ policy structures, which provide rewards only for voluntary reductions in emissions. Rather, in estimating responses to a carbon incentive, these studies typically examine a scenario that values all changes in forest carbon (i.e., both decreases and increases in emissions). This scenario reflects the incentives under a symmetric subsidy/tax system or cap-and-trade system that includes the forest sector, since these approaches entail economic rewards or penalties for all emissions changes. However, actual policy incentives for REDD+ are likely to be one sided, with compensation for voluntary reductions below a reference level but no penalty for increases in emissions above that level.

The selection of an appropriate reference level raises questions about whether the reductions being credited represent real and additional reductions (i.e., that would not have occurred without the policy) as well as concerns about distributional impacts (see Angelsen 2008). There are also efficiency concerns because voluntary participation decisions, and the associated leakage of emissions, are endogenous. This means that actors may not only exploit information asymmetries to capture rents but also opt out of programs if incentives for reductions are too low, possibly preventing the program from capturing cost-effective potential to reduce emissions and avoid leakage. Busch et al. (2009) analyze national decisions to opt voluntarily into a global deforestation reduction program under crediting reference levels that are based on historical deforestation emissions, as well as other proposed incentive designs. Although the differences in global emissions reductions are relatively small across scenarios, net global emissions reductions per unit cost are maximized by balancing incentives for reducing emissions in countries that have high historical deforestation rates with incentives not to increase emissions in areas that have historically low deforestation rates.

While voluntary participation is a general issue for crediting offsets from any source, a distinguishing feature of REDD+ is the focus on crediting emission reductions that occur on national or subnational scales. This approach has important advantages such as netting out leakage within the jurisdiction and mitigating adverse selection (e.g., Richards and Andersson 2001). However, it also creates implementation issues, including ensuring consistency of reference levels across spatial scales (Cattaneo 2011). In a study of potential REDD+ incentive structures in Indonesia, Busch et al. (2012) find that a more traditional site-by-site crediting approach creates the risk of paying some participants more than is necessary while providing insufficient incentives to others, and producing an overall shortfall for an administrator with a fixed budget. Aggregating the accounting to a larger geographic scale and sharing revenues and costs (for emitting above a reference level) among national and subnational jurisdictions reduces risks of budgetary losses and increases total emissions reductions closer to the level that would be achieved under a symmetric subsidy/tax or cap-and-trade system.

Carbon Prices over Time

A final issue in estimating potential REDD+ supply is whether carbon prices are expected to be constant or rising over time. The sectoral model estimates shown in Figure 1 assumed constant carbon price trajectories over time. Constant carbon prices could be a good proxy for a constant subsidy/tax or a carbon market with annual limits on the use of forestry mitigation, or offsets in general. However, many modeled climate policies project rising rather than constant carbon prices, with prices rising at the rate of interest, which would be the theoretical market outcome from eliminating intertemporal arbitrage opportunities. Rising carbon price pathways could result in significantly lower forestry mitigation in the near term because landowners would have incentives to delay afforestation and reforestation and other carbon accumulation activities to take advantage of higher future carbon prices (see, e.g., Sohngen and Sedjo 2006; van't Veld and Plantinga 2005). In these cases, supply estimates based on rising carbon price paths would be most appropriate. Thus the policy context is important when interpreting results from different REDD+ supply estimates.

Implications for Climate Management

The issues discussed here qualify the findings of most studies that estimate the role of REDD+ in GHG management for achieving climate or emissions objectives. Like the REDD+ supply studies, these studies have generally abstracted from the realities of climate policy and carbon market design and implementation. Incomplete and delayed participation, in terms of both countries and activities, reduces mitigation supplies and causes international leakage, which will result in more costly global mitigation (e.g., Calvin et al. 2009). In addition, these studies have not addressed the issue of temporal leakage, where forest carbon policy expectations could lead to accelerated deforestation (Rose and Sohngen 2011). Finally, most of these studies do not account for implementation and investor costs, uncertainties and other practical barriers, and policy design details, including the feasibility of establishing long-term expectations and using a credit bank over forty or a hundred years (see US EPA 2010 and Bosetti et al. 2011, respectively). It is important to note that these real-world complexities could also affect the economic modeling of emissions reduction and climate management potential in other economic sectors such as energy and industry.[10] In summary, the ultimate costs and net benefits of REDD+ alone and relative to other emission reduction options will depend on the specific policy designs

[10]For instance, Rose et al. (2013) quantify the relative risks across energy, land use, and other mitigation technologies, with forest carbon activities characterized as relatively high risk, as are other technologies, including geologic carbon dioxide sequestration.

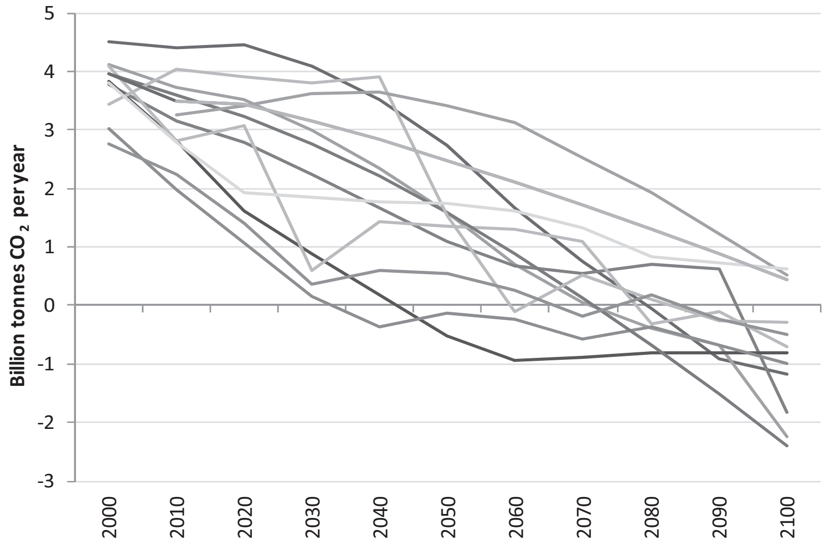

Figure 2 Recent global baseline land use change and forestry emissions projections from partial equilibrium and integrated assessment models ($GtCO_2$/year).
Note: For each model, the most recent projection and reported sensitivities are used.
Source: Developed from Fisher et al. (2007) and Clarke et al. (2009).

adopted across countries, sectors, and time, which have received relatively little economic analysis attention to date.

Research Needs and Challenges

Although economic modeling has yielded insights concerning REDD+ opportunities and policy design, there are numerous uncertainties that affect economic evaluation and the interpretation and application of the results for policymaking. As shown in Figure 2, there is a large variation in projected baseline global land use change and forestry emissions, which has implications for estimated forestry abatement potential over the next decades and century. The studies in Figure 2 project cropland expansion to be a key driver of future deforestation. However, these projections depend critically on uncertain assumptions regarding historical activity, biophysical characteristics, and projected population and economic growth, as well as agricultural productivity changes, which determine food demand and production opportunities and thus cropland expansion. Uncertainties in data and parameters complicate not only the modeling of REDD+ potential but also the design of international agreements and their implementation locally.[11] This section discusses these uncertainties and the limitations in modeling capability and application in order to provide insights for the interpretation and use of past research on REDD+ and to help identify opportunities for future research.

[11]For an empirical investigation of the effectiveness of national policies on deforestation see Pfaff, Amacher, and Sills (2013).

Uncertainties in Key Data and Parameters

Data and parametric uncertainties pose a significant challenge to projecting future emissions from land use as well as the costs of REDD+. Estimates of REDD+ costs and carbon sequestration potential are based on our understanding of carbon stocks, historical deforestation, and broader land use trends. Higher assumed carbon stocks (carbon per hectare) and/or lower deforestation pressures imply, all else equal, a lower marginal cost of reducing forest emissions, although lower estimated deforestation also limits the quantities of emissions that potentially could be reduced through REDD+. However, our basic understanding of historical deforestation rates, tropical land use transitions, and current carbon stocks is still relatively poor for many regions. These uncertainties are separate from uncertainties about future economic growth, population, technologies, and other factors, including potential climate feedbacks that could affect risks of disturbances and the productivity of agriculture and forestry (which few studies have addressed). Together these factors imply that we do not yet know the full extent of future uncertainties.

Uncertainties of terrestrial components

Terrestrial components, including carbon stocks and fluxes in forests and soils, are still the most uncertain elements of the global carbon cycle (Intergovernmental Panel on Climate Change 2007; Pan et al. 2011). Henry et al. (2011) estimate orders of magnitude ranges for aboveground forest carbon stocks and deforestation losses in Africa, although Saatchi et al. (2011) suggest less uncertainty concerning tropical forests. Estimates of soil carbon in other land cover types (e.g., cropland, pasture, savannah, grassland, and scrubland) are more uncertain, which has implications for the net climate benefits of land conversion. Uncertainties concerning the magnitudes and processes of degradation are even greater.[12]

Historical patterns of land use conversion

Historical land use conversion patterns are also not well understood. Consistent land conversion data for multiple time periods and countries are sparse; data on their relationship to economic drivers are even scarcer. To project land use emission changes, we need to know how readily forests and other land can be converted from one use to another (i.e., the elasticity of transformation). Top-down models have typically assumed that landowners are willing to convert land in response to changing economic incentives (e.g., Alig et al. 1997). However, statistical analyses have revealed that landowners are more reluctant to convert land (e.g., Lubowski, Plantinga, and Stavins 2006), suggesting higher forest carbon sequestration costs. Possible explanations for this difference include unobserved factors affecting the returns to different land uses, decision-making inertia, and option values associated with delaying decisions under uncertainty. Unfortunately, because few empirical analyses are available, global models have often relied on estimates for the United States.

[12]Emissions from degradation are estimated to account for at least 20 percent of global tropical forest emissions and up to 50 percent or more of total tropical forest emissions in Asia and Africa (see Griscom, Ganz, et al. 2009).

Changes in agricultural productivity

A critical uncertain parameter for projecting future deforestation is the expected rate of change in agricultural productivity. Holding productivity for crops and livestock at current levels greatly increases projected deforestation (Choi et al. 2011; Wise et al. 2009). However, total factor productivity improvements could lead to increased deforestation, especially if there is regional heterogeneity in productivity improvements, which is likely (Rose, Golub, and Sohngen forthcoming). Two other parameters that are important for projecting deforestation but are uncertain are the potential for increasing yields on existing cropland in response to crop prices and the productivity of new lands brought into agricultural production (Hertel, Tyner, and Birur 2010). For example, if we assume the marginal productivity of converted land is high, land is easily substituted by nonland inputs, and land conversion is expensive, we could find that the estimated land use change response to an increase in US corn ethanol production is an order of magnitude less global cropland area.[13] For REDD+, these same assumptions imply lower deforestation, lower REDD+ costs, and less regional leakage. The global supply and demand elasticities for agricultural commodities are also uncertain parameters (e.g., Roberts and Schlenker forthcoming).

Modeling Capability and Application

The current state of our modeling capability also poses challenges to projecting future emissions from land use and the cost effectiveness of REDD+. Our land use modeling capability is relatively limited, especially within general equilibrium models and IAMs (Fisher et al. 2007; Rose et al. 2012). Modeling land use competition endogenously within these frameworks is relatively new. Models have also tended to estimate forestry mitigation potential in isolation and are only beginning to analyze interactions between different sources of forestry abatement and other land-based abatement, as well as interactions with other parts of the economy, such as the energy sector. Meanwhile the dynamics of forestry and terrestrial carbon stocks and their investment implications are even more challenging to model (Sohngen et al. 2009), as is modeling access of unmanaged lands (Gurgel, Reilly, and Paltsev 2007; Rose, Golub, and Sohngen forthcoming). Although patterns of trade may change under REDD+ policies, economic models typically hold the distribution of trade fixed, an assumption consistent with short-term responses. This assumption has implications for estimating agricultural and forestry production and, in turn, the opportunity costs of deforestation and other land uses.

Role of forest degradation

Economic land use and forestry models have also largely ignored the role of forest degradation.[14] Degradation activities, such as selective logging, can catalyze deforestation by increasing access and risks of fire and subsequent degradation (Griscom, Ganz, et al. 2009). By ignoring degradation and these types of interactions, models could be underestimating the potential for tropical forestry mitigation, as well as mischaracterizing the processes and costs of REDD+ and the possibilities for leakage into degradation activities.

[13]See the online supplementary materials for an example.
[14]An exception is the Mosnier et al. (2011) analysis of deforestation in the Congo Basin.

Water availability and climate impacts

Analysis of the interactions between land use and water availability is also an incipient area for model development, as are the net climate impacts of land use (e.g., Betts et al. 2006) and potential climate change feedbacks including potentially large-scale drying and burning (i.e., dieback) of forests in the Amazon and possibly other regions (e.g., Vergara and Scholz 2011). Tropospheric ozone is another factor to consider as increased ground-level ozone levels due to air pollution will moderate potential increases in plant growth from higher atmospheric CO_2 concentrations (Felzer et al. 2005).

Societal benefits

Finally, although economic analysis of REDD+ has focused on carbon, tropical forests also provide other societal benefits—environmental services, biodiversity, aesthetic value, habitat, local goods, and recreation. Research suggests significant co-benefits for biodiversity from REDD+ incentive programs based on carbon (Busch et al. 2010; Strassburg et al. 2012), although shifting of agriculture to low-carbon yet high-biodiversity ecosystems is a risk (Miles and Kapos 2008). Payments for biodiversity plus carbon could significantly increase joint benefits at relatively low cost (Siikamäki and Newbold 2012; Venter et al. 2009), and they might even increase carbon benefits if rents to producers can be redirected (Busch forthcoming). Thus future research on REDD+ policy design should consider the synergies and potential tradeoffs between REDD+ and other forest benefits.

Summary and Conclusions

Our review of the economics literature concerning REDD+ finds that modeling efforts to date provide REDD+ insights regarding potential locations for investments, costs of supply, cost-effective deployment in helping achieve climate policy objectives, implications across economic sectors, and the implications of policy design. In particular, the literature indicates there are opportunities for significant low-cost climate change mitigation from both reducing deforestation emissions and enhancing forest carbon stocks, which could reduce the overall cost of achieving climate policy goals. Studies also suggest that if available in the near term, REDD+, particularly reduced deforestation, could provide the added benefits of dampening carbon market price volatility and serving as a hedging strategy against future uncertainties. However, these analyses have assumed unrealistic, idealized implementation conditions. Implementation challenges include establishing reference levels for crediting, designing voluntary incentives, investment risks, data uncertainties, and fragmentation and delays in the development and coordination of global forest carbon policies. Recent analyses that consider some of these realities of policy design and implementation suggest a more limited and nuanced mitigation role for REDD+. These real-world complexities, as well as numerous modeling challenges, suggest that REDD+ supply, cost savings, and net climate benefits are uncertain and will be highly dependent on policy and implementation features. Although most previous studies have excluded degradation (a potentially large emissions source), we conclude that because of the implementation and policy design uncertainties and challenges, estimates of

REDD+ potential that have relied on idealized assumptions may be biased high, especially for REDD+ in the near future.

It is important to note that while REDD+ presents some unique implementation issues, REDD+ activities are not unique in facing implementation challenges and uncertainties that can affect economic modeling results. Many mitigation activities confront uncertainties and investment risks, and policy design challenges such as data deficiencies, potential emissions leakage, adverse selection, and implications for other social priorities. Economists are just beginning to evaluate these issues. Research on and experience with REDD+ implementation will likely provide insights for designing large-scale incentive programs to reduce emissions in other sectors as well.

There are many research needs and opportunities for analyzing REDD+ policy designs at the global, national, and subnational levels including examining land use planning and other applications for ongoing REDD+ policy processes (e.g., Sathaye, Andrasko, and Chan 2011). There are also fundamental gaps and uncertainties in our understanding of historical conditions and trends and our ability to model future scenarios of land use changes, carbon fluxes, and relationships with prices and key economic drivers. Empirical analyses are needed to strengthen our modeling foundations, particularly as new remote sensing and other data become available. Projection and spatially detailed models will also be important for policy implementation. Sectoral and economy-wide simulation models are only beginning to integrate explicit land competition and allocation, and interactions between different forestry activities and across other land-based mitigation alternatives in agriculture and bioenergy. Important directions for future modeling include evaluating ways to achieve emission reductions while minimizing leakage across regions and REDD+ activities, alternatives for transmitting economic and financial incentives for REDD+ (and for managing associated risks and uncertainties), linkages between REDD+ activities and other social priorities (food security, development, climate adaptation, biodiversity, ecosystem services, water availability, agricultural policies), and tighter integration of forestry and other land-based activities with other parts of the economy.

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